# Energy Shocks and the Green Adoption Channel in a Small Open Economy

Guillaume, Boris Chafwehe, Francesca Diluiso

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**Abstract**

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of Auclert et al. (2024). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks under the two fiscal responses studied in the recent literature: a retail price cap (Langot et al., 2026) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption (Bayer et al., 2026). The price cap stabilises output but mechanically eliminates the green adoption margin, while the transfer delivers comparable stabilisation and leaves adoption intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. The transition loss is linear in the cap's intensity under a price shock and convex under a supply shock, and the *sign* of the cap's welfare effect flips with the elasticity of energy supply, reproducing the sign obtained by Langot et al. (2026) and Bayer et al. (2026) in their respective closures. This isolates *one* axis of their disagreement—the supply elasticity—without nesting the labour-cost and cross-border channels that also separate them. The policy *ordering* is robust to calibration; the *magnitude* of the adoption channel, and even the sign of its output effect under a price shock, depend on the calibrated switching cost and taste scale, which we report and discuss.

## 1 Introduction

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. Auclert et al. (2024) study a small open economy and stress the negative externality when all importers subsidise simultaneously. Bayer et al. (2026) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. Langot et al. (2026) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well. All three treat the household's energy technology as fixed.

We relax exactly that margin. Households can adopt a green durable that insulates them from the fossil price, at a resource cost. The energy shock raises the return to adoption, so the crisis itself can accelerate the transition—unless policy removes the price signal that drives it. Our energy-market closure spans one axis of the disagreement between Bayer et al. (2026) and Langot et al. (2026) in a single parameter: with $\varepsilon^E = \infty$ the economy is a price taker (elastic supply, as in Langot); with $\varepsilon^E$ finite the quantity is fixed and the world price clears the market (as in Bayer). Both closures are reported below. We stress that this spans the supply-elasticity dimension only. Langot et al. (2026)'s shield also operates through a labour-cost channel—by containing the cost of labour it supports employment—and Bayer et al. (2026)'s central result concerns cross-border spillovers within a monetary union; a small open economy with an exogenous foreign block reproduces neither.

## 2 Model

### 2.1 Environment

The aggregate structure is Auclert et al. (2024): a small open economy with sticky wages, a CES consumption basket nesting energy against a home/foreign bundle, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share $\zeta^E$ of the energy endowment. We build the model literally on the ARS aggregate blocks and replace its representative, homothetic household with a heterogeneous-agent household that carries a discrete durable state; everything else in the aggregate economy is inherited unchanged.

### 2.2 Model at a glance: the maintained assumptions

Because the household block departs from ARS in several coupled ways, we list the maintained assumptions explicitly before deriving anything. A reader who accepts this list can reconstruct every result.

1. **One new margin.** The only structural addition to Auclert et al. (2024) is a discrete brown/green durable adoption choice. Conditional on the durable state, preferences are ARS's: the same CES energy nest ($\alpha_E$, $\eta_E$) and the same home/foreign nest. There is no within-durable-type energy-share heterogeneity.
2. **Green insulates from the fossil price.** A green adopter buys energy at $P_G^E$, which in the baseline ($\rho_g = 0$) is constant and does *not* respond to the fossil shock. Adoption is therefore, in the baseline, full insulation—an upper bound on the channel.
3. **Green is resource-cheaper, not just household-cheaper.** The energy an adopter uses is assumed to have a genuinely lower resource cost (a domestic low-marginal-cost technology such as rooftop solar or a heat pump). This is what licenses booking the adopters' energy saving as reduced imports (Section 2.8); we do not model the technology that delivers it (no green production sector, no green investment).
4. **The switching cost is a resource cost booked externally.** Adopting costs $\psi_g$ units, booked in the current account as an import, by accounting necessity under ARS's markup/capitalised-profits convention (Section 2.8).
5. **Unit of account is the domestic good.** We set $p^{num} = p_H$ (Section 2.3). This is a measurement choice, orthogonal to the nominal anchor; it keeps the resource constraint independent of how the CPI weights brown and green energy.
6. **Energy supply is a single dial.** $\varepsilon^E = \infty$ gives the price-taking closure (exogenous world price, price shock); $\varepsilon^E$ finite fixes the quantity and lets the world price clear (supply shock). The steady state is identical under both.
7. **Monetary baseline is the constant-real-rate rule** of Auclert et al. (2024). Under this rule the real exchange rate is pinned, so a domestic transfer has no terms-of-trade effect; we also report a Taylor rule.
8. **First order throughout.** All impulse responses are first-order (sequence-space Jacobian). The steady state is common to every experiment, since both policy instruments are zero at the steady state.

### 2.3 Unit of account

The household block is numeraire-generic: it reads a price of the unit of account relative to the CPI, $p^{num}$, a real return $1 + r^{num} = (1 + r)\, p^{num}_{-1}/p^{num}$, and after-tax labour income $atw^{num} = atw/p^{num}$, and returns asset holdings converted by $A^{CPI} = A\, p^{num}$ before meeting

CPI-denominated objects in asset market clearing and the current account. We set $p^{num} = p_H$, the domestic good. This makes the resource constraint independent of how the CPI weights brown and green energy, which matters once the two coexist (Section 2.7). Everything contractually written in CPI units—fiscal transfers, the ARS subsistence term, and the switching cost $\psi_g$—is divided by $p^{num}$ on the way into the budget.

Two remarks make the convention transparent. First, the choice of numeraire is the choice of nominal anchor, not of deflator: the Fisher equation is written once in CPI terms ($1 + r^{ante} = (1 + i)/(1 + \pi)$), and $p^{num} = p_H$ then delivers, after the relative-price conversion, a household discount factor deflated by *core* (domestic-good) inflation, $1 + r^{num} = (1 + i_{-1})/(1 + \pi^H)$. Second, output $y$ is a physical quantity of the domestic good and is therefore already expressed in numeraire units—no re-deflation is needed to read it as a real quantity; CPI-deflated GDP would be $p_{H/P}\, y$, and the two differ only by the (energy-driven) movement in $p_{H/P}$.

### 2.4 The durable state

Each household carries a durable state $d$, the *pair* (holding entering the period, holding chosen last period), needed to charge the switching cost:

$$d \in \{BB,\ BG,\ GB,\ GG\},$$

where $GB$ (green now, brown before) is the state that pays $\psi_g$. Brown is absorbing; green breaks down to brown at rate $\delta_g$. The state $BG$ carries zero mass by construction, so the effective state is three-point. Stages, in forward order, are `[dep, prod, durables, consav]`: breakdown, idiosyncratic productivity, the discrete adoption choice (a logit with taste scale $\sigma_\varepsilon$), and the EGM consumption–savings step. The adoption stage uses the discrete-continuous structure of Iskhakov et al. (2017).

Three subtleties are worth making explicit, because the calibration and the sign results depend on them. (i) The choice compares continuation values across durable states; the fiscal transfer enters cash-on-hand *identically* across durable states, so it shifts adoption only through the curvature of the value function, which is why the transfer barely moves the green share. (ii) The asymmetry "brown absorbing, green depreciating at $\delta_g$" means green ownership requires *active* re-adoption each time the durable breaks down, so a permanent green share is sustained by a flow of switchers $D^{sw} = \delta_g D^G$ in steady state. (iii) The logit taste scale $\sigma_\varepsilon$ smooths the choice: at exact indifference the model assigns a structurally fixed switching probability (not one half), a consequence of the asymmetric choice sets across states. The magnitude of the adoption response is governed jointly by $\psi_g$ and $\sigma_\varepsilon$; see Section 7.

### 2.5 Energy prices

We distinguish three prices, all relative to the CPI:

| | |
|---|---|
| $p^E$ | wholesale/retail market price of brown energy, before any subsidy |
| $P^E_B = (1 - \tau^E)\, p^E + \tau^E \bar{p}^E$ | price paid by a *brown* household |
| $P^E_G = \varrho\, \bar{p}^E \left(P^E_B / \bar{P}^E_B\right)^{\rho_g}$ | price paid by a *green* adopter |

with $\varrho$ the steady-state green/brown ratio and $\rho_g$ the pass-through of the fossil shock to the green price. In the baseline $\rho_g = 0$, so $P^E_G$ is constant: adopters are fully insulated.

### 2.6 How the durable enters the budget

Households of different durable types face different energy prices, hence different *consumption bundle prices*. We give each type its own exact CES cost-of-living index, reusing ARS's own energy-nest parameters $(\alpha_E, \eta_E)$:

$$p_d^E = P_B^E + \mathbb{I}[d \text{ green}]\,(P_G^E - P_B^E), \qquad p_d = \Big[1 + \alpha_E\big((p_d^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}\big)\Big]^{\frac{1}{1-\eta_E}},$$

and the budget constraint is $(p_d/p^{num})\,c + a' = \mathrm{coh}(d)$. The second expression is exact given the CES identity enforced by the equilibrium block, so brown households have $p_d \equiv 1$ at every date. With type-specific prices, aggregate energy demand is $\sum_d c_E(d)$, not the CES demand evaluated at the aggregate index—the two differ by Jensen's inequality (0.5% here)—so aggregate $c_H$, $c_F$, $c_E$ are built from the household block's own heterogeneous-agent outputs.

## 2.7 What the CPI measures

With two energy prices in the economy, the energy leg of the CPI could be anchored on the brown price alone or on the share-weighted index $[(1 - D^G)(P_B^E)^{1-\eta_E} + D^G(P_G^E)^{1-\eta_E}]$. We anchor on the brown price: it is what statistical agencies do (the energy basket is not reweighted quarterly for technology adoption), it preserves $p_d \equiv 1$ for brown households exactly, and feeding $D^G$ back into the CPI would close a DAG cycle requiring the CPI to be promoted to a model unknown. Because the unit of account is the domestic good, this choice does not touch the resource constraint; it affects only the measured deflator and, through it, the monetary rule. The gap between the share-weighted and brown-anchored CPI has closed form

$$\phi^{1-\eta_E} = 1 + \alpha_E\, D^G\Big[(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}\Big],$$

computed ex post outside the model. On the price shock without policy it reaches 0.14 annualised percentage points, about 1.5% of the amplitude of measured CPI inflation (0.23 pp, 2.3%, on the supply shock; Figure 1).

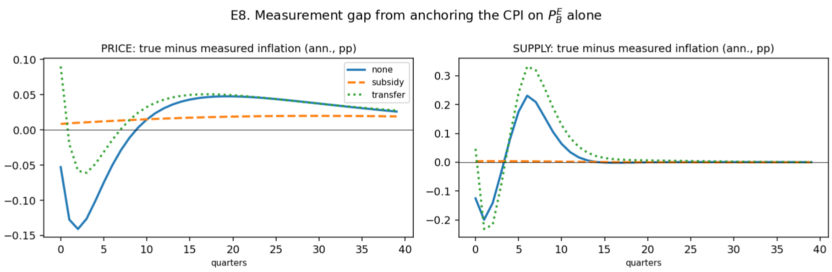


Figure 1: Measurement gap from anchoring the CPI on $P_B^E$ alone: true minus measured annualised inflation, computed ex post.

## 2.8 Balance of payments

Two durable-margin terms enter the current account. The **switching cost** is booked as an import, $\text{imports}_{dur} = \psi_g D^{sw}$: booking it as domestic demand instead fails here, since with $\mu_{ss} > 1$ and capitalised profits, extra demand for the home good creates a marginal dividend owned by nobody. The **energy saving** of adopters, $(P_B^E - P_G^E)\,C_E^G$, is booked as a real import saving: this asserts that cheaper energy for adopters is a genuine resource saving for the country, not a domestic transfer, and gives the windfall an external counterpart consistent with asset market clearing.

**Maintained assumption on green energy.** Booking the adopters' energy saving as reduced imports is correct only if green energy is genuinely cheaper *in resource terms*, not merely cheaper to the household. We maintain this: once the green durable is installed, its energy input (rooftop solar, a heat pump drawing on low-marginal-cost electricity) has a lower resource cost than imported fossil energy, so an adopter's switch is a real reduction in the country's energy import bill rather than a domestic transfer. We do not model the origin of that saving—there is no green production sector and no green investment; the steady-state ratio $\varrho$ and the zero pass-through $\rho_g$ stand in for a fully insulated, resource-cheaper technology, which makes the adoption benefit an upper bound (Section 7). Under this assumption the current-account entry is an accounting identity with a genuine external counterpart, not a free lunch.

### 2.9 Policy instruments

Both instruments are the extreme (full-compensation) versions, as in Bayer et al. (2026).

$$\text{Price cap:} \quad P^E_{B,t} = (1-\tau^E)p^E_t + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow \text{retail price fixed at pre-crisis level} \tag{1}$$

$$\text{Slutsky transfer:} \quad tr_{it} = \iota\,(p^E_t - \bar{p}^E)\,\bar{c}_{E,i}, \qquad \iota = 1 \Rightarrow \text{full compensation on } \textit{pre-crisis} \text{ use} \tag{2}$$

Both are debt financed and repaid through the distortionary labour tax. The transfer base $\bar{c}_{E,i}$ is each household's steady-state energy demand. The cap operates directly on $P^E_B$, exactly the price entering the brown–green gap, and therefore compresses the adoption incentive; the transfer is lump-sum with respect to current choices and leaves the relative price intact.

## 3 Calibration

| Parameter | Value | Source / target |
|---|---|---|
| $\alpha_E$ energy expenditure share | 0.04 | Auclert et al. (2024) |
| $\eta_E$ energy substitution | 0.10 | Auclert et al. (2024) |
| $r$, $\mu_{ss}$ | 0.01, 1.03 | Auclert et al. (2024) |
| $\beta$ heterogeneity (3 types) | spread 0.06 | MPC $\approx 0.30$ |
| $\delta_g$ green breakdown rate | 0.05 | assumption (5 yr life) |
| $\varrho = P^E_G/P^E_B$ steady-state gap | 0.80 | assumption |
| $\psi_g$ switching cost | 0.538 | targets $D^G_{ss} = 0.05$ |
| $\sigma_\varepsilon$ taste scale | 0.05 | calibrated jointly with $\psi_g$ |
| $\rho_g$ pass-through to green price | 0 | upper bound on the channel |

Table 1: Calibration. The steady state is identical across all experiments. $\psi_g$ and $\sigma_\varepsilon$ are calibrated jointly to the level target $D^G_{ss} = 0.05$; $\sigma_\varepsilon$ is not separately pinned down by an adoption-elasticity moment (Section 7).

## 4 Experiments and results

The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life, as in Auclert et al. (2024). The *supply* shock is a 5% cut in available energy for 6 quarters, after which the world price clears the market endogenously. Impulse responses are reported as 100× the level deviation from steady state. Variables whose steady state is one (prices, $w$, $n$, and the population shares $D^G$, $D^{sw}$) read directly as percent or percentage-point deviations, and $C$ (steady state $\approx 0.997$) is within rounding of a percent; $y$ and gdp have steady state $\approx 0.985$, so their entries are level deviations ×100, within about 1.5% of the exact percentage. Energy quantities are a share of the energy steady state ($\approx 0.040$).

| Shock | Policy | $y(0)\%$ | $\sum y$ | $\pi(0)$ ann. | peak $D^G$ pp | fiscal cost | adoption $\to \sum y$ |
|---|---|---|---|---|---|---|---|
| price | none | $-1.12$ | $-30.1$ | 9.14 | **9.18** | 0.00 | $-1.60$ |
| price | cap | $-0.00$ | $-20.9$ | $-1.53$ | **0.01** | 53.82 | $+0.36$ |
| price | transfer | $+1.20$ | $+12.7$ | 11.08 | **9.32** | 53.39 | $-1.74$ |
| supply | none | $-0.71$ | $-8.16$ | 10.06 | **2.33** | 0.00 | $+0.38$ |
| supply | cap | $-0.74$ | $-46.9$ | $-0.96$ | **0.01** | 26.09 | $+0.18$ |
| supply | transfer | $+1.33$ | $+2.72$ | 14.46 | **3.39** | 26.56 | $+2.48$ |

Table 2: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin: the difference between the model with the margin open and the same model with it shut.

### 4.1 The crisis moves the transition, in a shock-dependent direction

With no policy, the green share rises by up to 9.2 percentage points under the price shock (from 5.0% to 14.2% of households) and 2.3 pp under the supply shock (Figure 2). The output effect of the adoption margin, however, is not one-signed. Under the **supply** shock it is expansionary (+0.38 points of cumulative output with no policy, +2.48 under the transfer): adopters escape the energy bill and their real income falls less. Under the **price** shock it is contractionary (−1.60 points with no policy, −1.74 under the transfer): the switching cost is booked as an import in exactly the periods household income is falling hardest (Section 2.8), and at the calibrated $\psi_g$ this resource cost outweighs the energy-bill saving. The sign of the adoption channel's contribution to output is therefore not a general property of the model—it depends on the calibrated switching cost, on which shock is driving adoption, and (Section 7) on the accounting of $\psi_g$.

### 4.2 The price cap eliminates the adoption margin

Under the cap, the peak green share response falls to essentially zero under both shocks (from +9.18 to +0.01 pp on the price shock, from +2.33 to +0.01 on the supply shock): the adoption channel is not attenuated, it is switched off. Its contribution to cumulative output collapses to near zero accordingly (−1.60 → +0.36 on the price shock, +0.38 → +0.18 on the supply shock). The mechanism is direct: the cap holds $P_B^E$ at its pre-crisis level, and $P_B^E$ is precisely what determines the return to adoption.

### 4.3 The transfer preserves adoption, at a cost to output

The Slutsky transfer leaves the green share response essentially unchanged (9.32 vs. 9.18 pp under the price shock, 3.39 vs. 2.33 under the supply shock) while delivering more output stabilisation than the cap, at similar fiscal cost: subsidies destroy the transition margin, transfers do not. Leaving adoption intact is not free for output under the price shock, however: the transfer's cumulative-output gain is *smaller* with adoption open than shut (−1.74 points from the margin, Table 2), for the same resource-cost reason as above.

### 4.4 Fixed supply makes the cap destructive

Under the supply shock the cap yields cumulative output of −46.9 against −8.2 with no policy: with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief. This survives the addition of an adoption margin. It is also precisely the configuration in which Langot et al. (2026), assuming elastic supply, find the cap effective: our elastic-supply column agrees with them (−20.9 vs. −30.1 for cap vs. no policy under the price shock).

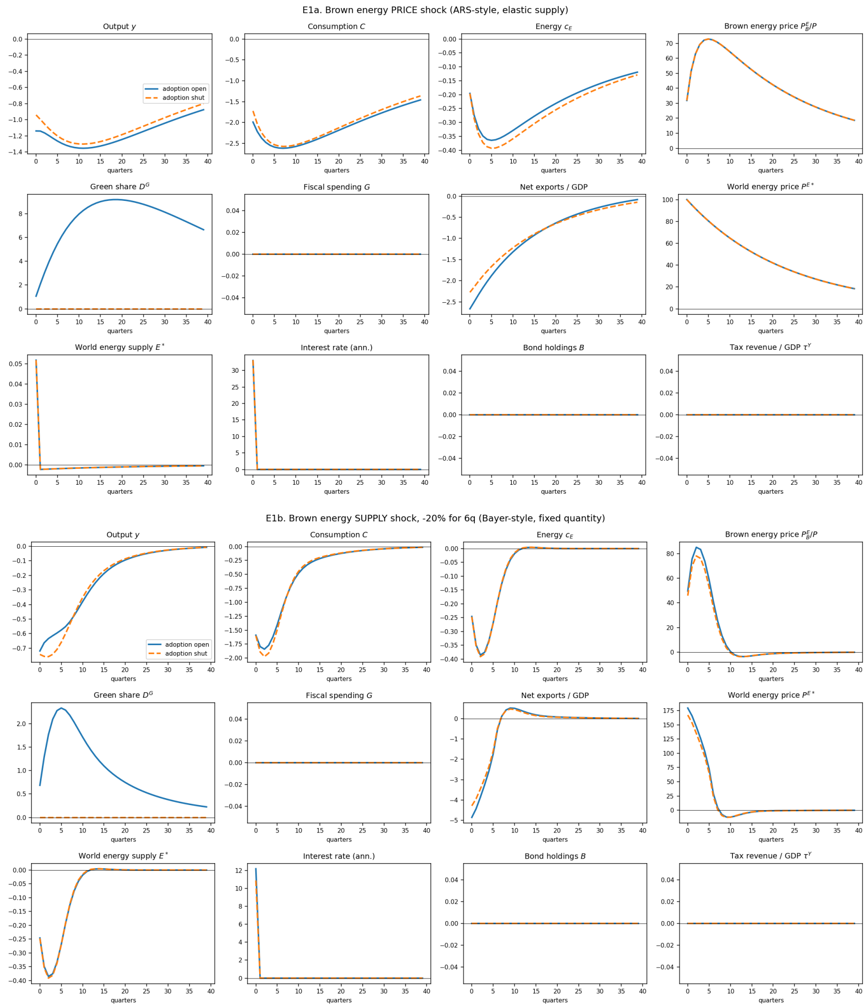


Figure 2: Brown energy shocks, adoption open (solid) vs. shut (dashed), no policy. Top: price shock, elastic energy supply. Bottom: supply shock, $-5\%$ for 6 quarters, fixed quantity.

### 4.5 The cap is a dial, not a switch

Table 2 used $\tau^E = 1$, a complete price freeze. Table 3 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in $\tau^E$ (half a cap costs a bit under half the transition: 4.53 against 9.18 pp). Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic—but, as the next result shows, much more damaging to everything else.

### 4.6 The welfare effect of the cap flips sign with the supply elasticity

This is the sharpest result in the paper. Under the price shock, tightening the cap *raises* welfare monotonically, from $-1.30\%$ to $-0.57\%$; under the supply shock it *lowers* welfare monotonically, from $-0.59\%$ to $-1.28\%$. The sign of the cap's welfare effect therefore hinges on the supply elasticity. Langot et al. (2026), assuming elastic supply, find the French shield effective; Bayer et al. (2026), assuming fixed union-wide supply, find subsidies destructive. Our two closures reproduce the *sign* each obtains, which locates one source of the contrast in a single parameter. We do not claim to reconcile the two papers fully: Langot's result also runs through a labour-cost channel and Bayer's through cross-border spillovers in a monetary union, and our small open economy nests neither. The narrower point we establish is that, once the supply elasticity is fixed,

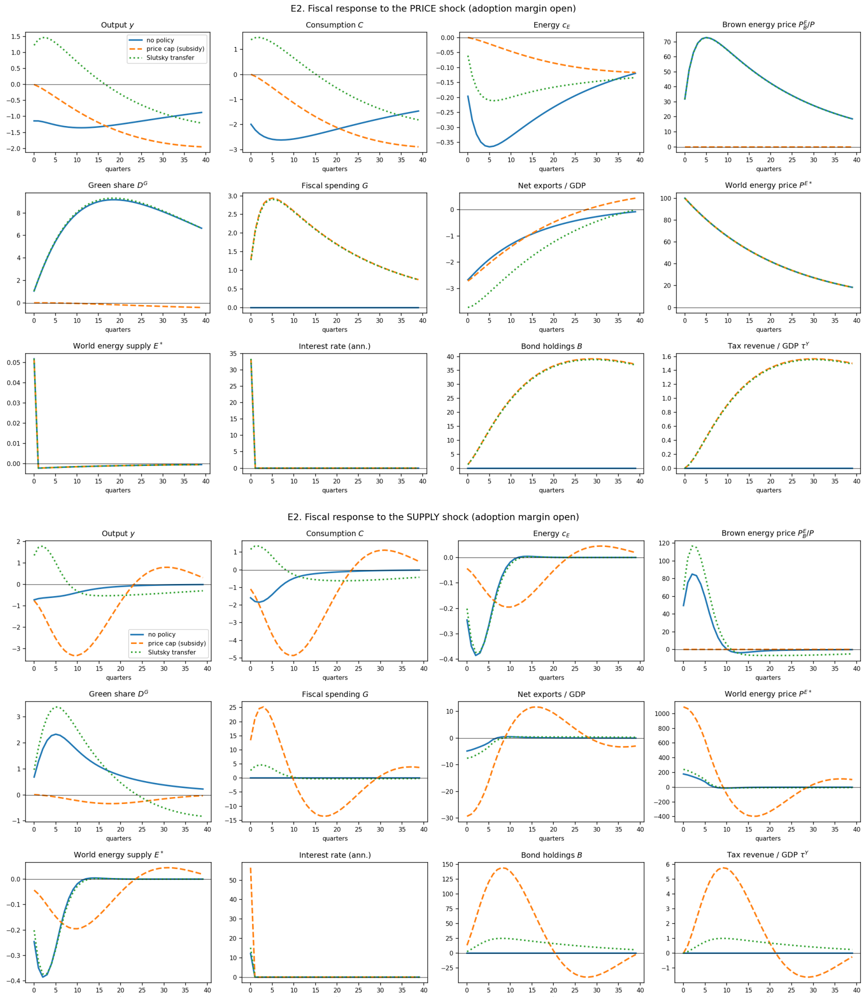


Figure 3: Fiscal response, adoption margin open: no policy (solid), price cap (dashed), Slutsky transfer (dotted). Top: price shock. Bottom: supply shock.

the direction of the cap's welfare effect is determined—so part of the empirical disagreement reduces to measuring that elasticity at the relevant horizon.

**Private welfare and the transition are distinct.** The CEV above is a *private* household-welfare measure: it values consumption and leisure along the transition and prices no climate or transition externality. Two results should therefore be kept separate. First, on private welfare alone the Slutsky transfer dominates the cap under *both* shocks ($-0.21$ vs. $-0.57$ under the price shock, $-0.14$ vs. $-1.28$ under the supply shock, Table 3); the case for the transfer over the cap does not rest on the transition. Second, and separately, the transfer preserves the green adoption margin that the cap eliminates (Section 4.3)—a positive statement about the energy transition, not a welfare ranking. We deliberately do not aggregate the two into a single social-welfare number: that would require a social value of adoption (a carbon price times avoided brown-energy use) that we do not model. A positive such value only reinforces the private ranking; it cannot reverse it here, since the transfer already dominates.

### 4.7 Distributional incidence

Without policy the impatient lose far more than the patient ($-2.10$ vs. $-0.14$, roughly $15\times$). The patient remain the least-hurt group throughout the cap sweep. One clear distributional result:

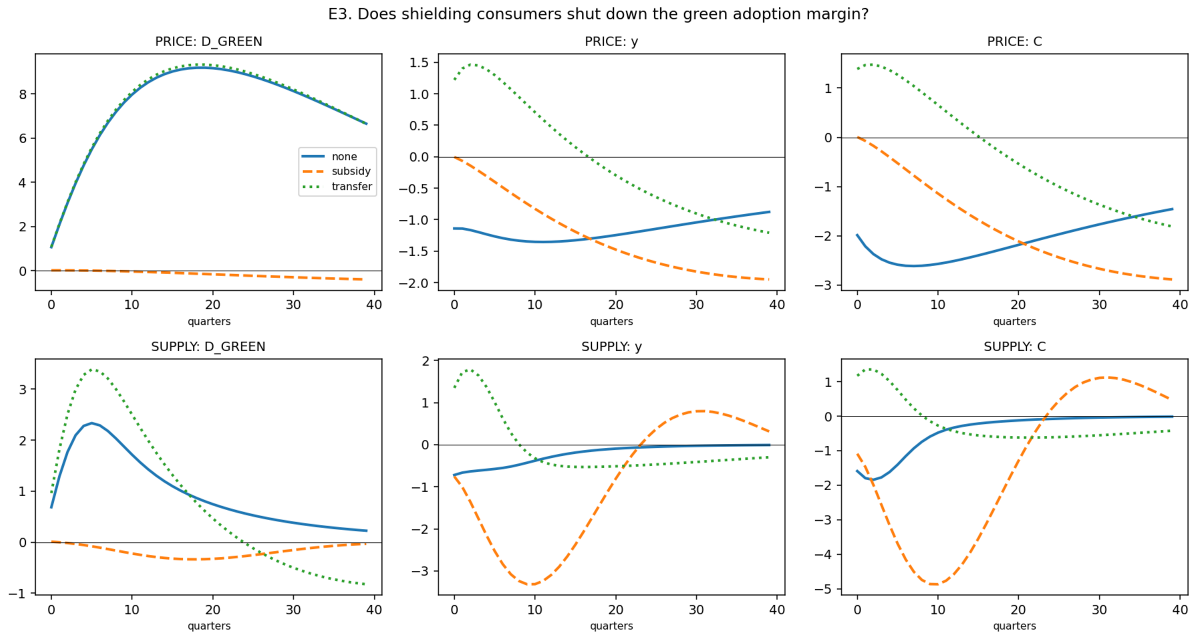


Figure 4: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

under the transfer, the **patient type's CEV turns positive** (+0.17%) while impatient and middle types remain negative. Indexed to pre-crisis energy use, the Slutsky compensation overcompensates low-MPC, low-energy-exposure households relative to their true loss—a symptom of the fiscal block's lack of investment margin to crowd out the transfer (Section 7).

# 5 Relation to the literature

This section is written for the coauthors: it states, paper by paper, what our setup changes relative to the reference and what that change does to the results. Compact one-page summaries of each reference are in Appendix A.

## 5.1 Relative to Auclert et al. (2024)

**Setup.** We inherit the entire ARS aggregate economy—the CES energy nest, sticky import prices, the mutual fund, the energy suppliers with intertemporal arbitrage, the real-wage-stabilising wage Phillips curve, and the constant-real-rate monetary baseline—and change the household in four coupled ways. (i) ARS's household is homothetic with a fixed energy technology: every household shares one CES basket and one CPI, and energy demand is $\alpha_E(P_E/P)^{-\eta_E}c$. We add a discrete brown/green durable state and a DC-EGM adoption choice, so the energy technology becomes an endogenous, household-level object. (ii) ARS has a single household energy price; we have two $(P^E_B, P^E_G)$ and a type-specific cost-of-living index. (iii) ARS uses the CPI as unit of account; we use the domestic good, so that the resource constraint is invariant to how the CPI weights brown and green energy. (iv) We add two current-account terms absent in ARS—the switching cost as an import, the adopters' energy saving as an import saving. The energy-supply closure is a generalisation of ARS: their baseline is our $\varepsilon^E = \infty$ price-taking case, and their coordinated world equilibrium (endogenous world price) is the two-country analogue of our $\varepsilon^E$-finite case.

| Shock | Policy | peak $D^G$ pp | $\sum y$ | $\pi(0)$ | fiscal | CEV % |
|---|---|---|---|---|---|---|
| price | cap $\tau^E = 0$ | 9.18 | $-30.1$ | 9.14 | 0.00 | $-1.30$ |
| price | cap $\tau^E = 0.25$ | 6.86 | $-27.8$ | 6.50 | 13.39 | $-1.12$ |
| price | cap $\tau^E = 0.50$ | 4.53 | $-25.5$ | 3.84 | 26.83 | $-0.94$ |
| price | cap $\tau^E = 0.75$ | 2.20 | $-23.2$ | 1.17 | 40.30 | $-0.76$ |
| price | cap $\tau^E = 1$ | 0.01 | $-20.9$ | $-1.53$ | 53.82 | $-0.57$ |
| price | transfer | 9.32 | $+12.7$ | 11.08 | 53.39 | $-0.21$ |
| supply | cap $\tau^E = 0$ | 2.33 | $-8.16$ | 10.06 | 0.00 | $-0.59$ |
| supply | cap $\tau^E = 0.25$ | 2.21 | $-10.60$ | 9.28 | 5.96 | $-0.63$ |
| supply | cap $\tau^E = 0.50$ | 1.99 | $-14.80$ | 8.07 | 15.34 | $-0.70$ |
| supply | cap $\tau^E = 0.75$ | 1.50 | $-23.27$ | 5.89 | 30.89 | $-0.84$ |
| supply | cap $\tau^E = 1$ | 0.01 | $-46.85$ | $-0.96$ | 26.09 | $-1.28$ |
| supply | transfer | 3.39 | $+2.72$ | 14.46 | 26.56 | $-0.14$ |

Table 3: Dose-response in the cap intensity. CEV is the consumption-equivalent variation relative to the steady state; negative means households would pay that share of permanent consumption to avoid the scenario. It is a *private* household-welfare measure and prices no climate externality; the same holds for the distributional CEVs below.

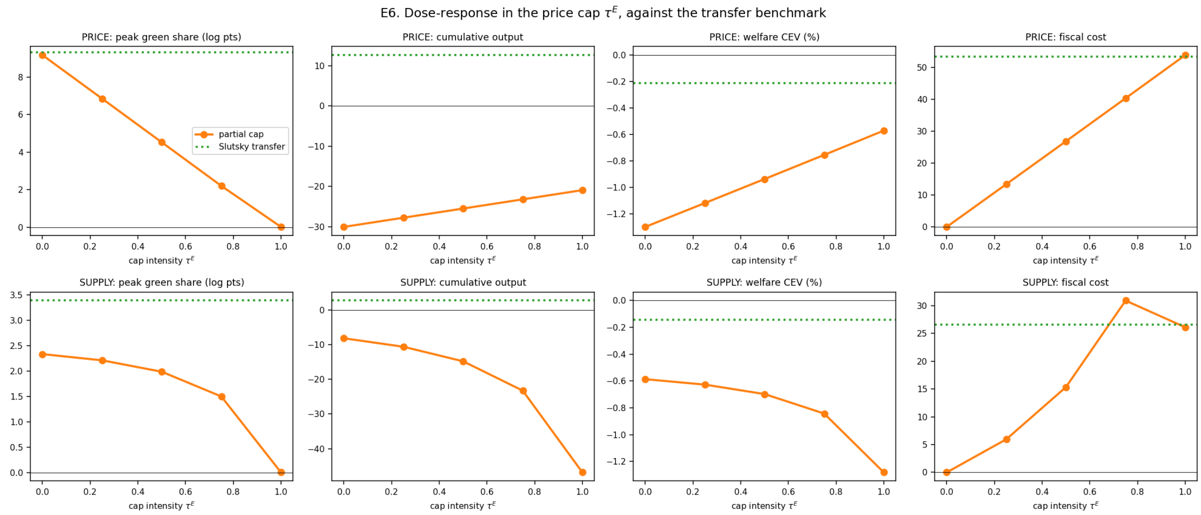


Figure 5: Dose-response in $\tau^E$ (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

**Results.** ARS's central mechanism—low $\chi$ makes the energy price shock contractionary through the real-income channel—is preserved and operates in the background of every one of our experiments. Where we depart is that ARS's fiscal ranking is entirely about output, inflation and inequality: in ARS the subsidy is a domestic "silver bullet" that stabilises all three, and its only drawback is a *cross-country* externality (coordinated subsidies spike the world price and become self-defeating). Our model adds an orthogonal cost of the same subsidy that ARS cannot see: capping $P_B^E$ eliminates the green adoption margin (peak $D^G$ from 9.18 to 0.01 pp), so the subsidy is not neutral for the transition. Two further connections are worth flagging to the coauthors. First, our supply-shock result—the cap is destructive under inelastic supply—is the *single-country* counterpart of ARS's coordination result: in both, capping the price under an endogenous/scarce energy price prevents the necessary substitution and deepens the recession. Second, our monetary baseline being ARS's constant-real-rate rule is what pins the real exchange rate, so a domestic transfer has no terms-of-trade effect; this is inherited, not assumed anew.

| Scenario (price shock) | CEV mean | impatient | middle | patient |
|---|---|---|---|---|
| no policy | −1.30 | −2.10 | −1.65 | −0.14 |
| cap $\tau^E = 0.5$ | −0.94 | −1.45 | −1.25 | −0.11 |
| cap $\tau^E = 1$ | −0.57 | −0.78 | −0.85 | −0.08 |
| transfer | −0.21 | −0.36 | −0.45 | +0.17 |

Table 4: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs. Private household welfare only.

### 5.2 Relative to Bayer et al. (2026)

**Setup.** Bayer et al. is a two-country HANK monetary union (Germany one-third, rest of the union two-thirds), with a fixed nominal exchange rate, built to study *spillovers*. Our economy is a single small open economy with an exogenous foreign block, so by construction we cannot represent their central object—the cross-border, beggar-thy-neighbour transmission of a national subsidy. Three further setup differences matter. (i) Supply: Bayer fix the union-wide energy *quantity* and let a common price clear; this is exactly our $\varepsilon^E$-finite closure, so our supply-shock experiments are the natural comparison. Our shock is smaller (a 5% cut for 6 quarters against their 20% cut for 6 quarters), chosen for first-order reliability. (ii) Supply side: Bayer has a rich medium-scale block—two assets (liquid bonds, illiquid capital), an investment margin with capacity utilisation, energy in production as well as consumption, and GHH preferences. We have a single asset, no capital, and energy in consumption only in the baseline. (iii) Heterogeneity: Bayer's headline distributional mechanism is *within-country energy-share heterogeneity*—poor households have systematically higher energy shares (a high/low-intensity Markov type correlated with income), so they suffer larger real-income losses. We assume homothetic energy demand conditional on the durable type, so we cannot speak to their energy-exposure distributional results; our heterogeneity is over discount factors and over the adoption margin instead. Finally, their monetary policy is a standard Taylor rule, not the constant-real-rate rule; they show this matters, because under a Taylor rule the subsidy also suppresses *intertemporal* substitution through household-specific real rates.

**Results.** Under the supply closure we reproduce the qualitative core of Bayer et al.: the Slutsky/Hicks transfer welfare-dominates the price cap (our private CEV −0.14 vs. −1.28), and capping the price under a fixed quantity is inefficient because it prevents the substitution the scarcity requires—our "cap pushes towards Leontief" is their "subsidising under a supply contraction is inefficient." What we add is the adoption margin, which Bayer et al. do not have: the same cap that is inefficient for output also switches off the transition, and the transfer that is welfare-superior also preserves it. What we *cannot* deliver is their spillover result: with no second country, we have nothing to say about whether one country's subsidy harms its neighbours. This is the reason we are careful (Section 4.6, and the abstract) not to claim that a single parameter "reconciles" Bayer and Langot; we reproduce the domestic sign, not the cross-border mechanism.

### 5.3 Relative to Langot et al. (2026)

**Setup.** Langot et al. study France as a single economy inside the euro area, with an ECB Taylor rule down-weighted for France's size. Two structural choices distinguish their model from ours. (i) They take the energy price as *exogenous* (fed from the government's official forecasts), so France is a price taker facing a perfectly elastic supply at that price—this is our $\varepsilon^E = \infty$ closure, and their price-shock scenario is the right comparison. (ii) Their energy substitution elasticity is $\eta_E = 0.5$, five times ours, and they add an *incompressible* (subsistence) level of energy consumption, which delivers non-homotheticity: poor households have a larger incompressible share, hence a higher effective energy share and a lower price elasticity. This is their distributional mechanism, analogous in spirit to Bayer's energy-share heterogeneity but generated by subsistence rather than by types. Our model buys energy substitution on a different margin: we keep the *intensive* elasticity low ($\eta_E = 0.1$, as in ARS) and obtain substitution from the *extensive* adoption margin instead. A third, methodological difference matters for how the results should be read: Langot et al. conduct an *ex-ante, conditional-forecast* evaluation—they back out the sequence of structural shocks that makes the HANK reproduce the French government's published forecasts for output, inflation and debt, then run counterfactual policies holding those shocks fixed. Ours (like ARS and Bayer) is a clean MIT-shock impulse-response exercise, not a fit to a specific historical episode.

**Results.** Under the elastic (price) closure we agree with Langot's headline: the cap cushions output ($\sum y = -20.9$ vs. $-30.1$ with no policy) and, on private welfare, tightening the cap raises the CEV ($-1.30 \rightarrow -0.57$)—consistent with their finding that the shield supports growth and curbs inflation. But two qualifications are important for the coauthors. First, Langot's pro-shield result rests substantially on a *labour-cost* channel: by containing the retail energy price the shield prevents the price–wage spiral, holds down the cost of labour, and thereby supports employment; in their comparison the shield beats both wage indexation and a targeted transfer precisely on this employment margin. Our wage block and our lack of an investment/employment margin mean we reproduce the *sign* of their result but not the full mechanism—hence we do not claim the Langot–Bayer contrast is "only" about the supply elasticity. Second, Langot et al. hold the energy technology fixed, so like ARS and Bayer they do not see the cap's cost to the transition, which is our contribution.

# 6 Solution invariance

Solving the model under the CPI numeraire instead of the domestic good gives a bit-identical steady state and impulse responses agreeing to $2 \times 10^{-5}$ on $y$ and $3 \times 10^{-5}$ on $C$ (about 0.15% of the respective amplitudes)—a relabeling of the same economy, not two different models.

The 5% supply cut used throughout is a deliberately conservative, illustrative magnitude. The results we emphasise—the policy *ordering* (transfer above cap), the elimination of the adoption margin under the cap, and the sign reversal of the cap's welfare effect across supply closures—are qualitative and hold across the range of shock sizes over which the model solves reliably.

# 7 Limitations

1. **The taste scale $\sigma_\varepsilon$ is calibrated, not estimated.** $\psi_g$ and $\sigma_\varepsilon$ jointly hit $D^G_{ss} = 0.05$, but one level moment cannot separately identify both a switching cost and a taste scale: a second moment (e.g. the semi-elasticity of adoption to a purchase subsidy) is needed and would discipline the sign found in Section 4.1 along with the magnitudes throughout. The *ordering* of the cap-vs-transfer results does not depend on it, since it follows from the relative price alone.

2. **The switching cost is booked as an import by accounting necessity.** Under ARS's $\mu_{ss} > 1$ with capitalised profits, booking $\psi_g D^{sw}$ as domestic demand creates a marginal dividend owned by nobody and breaks asset-market clearing; we therefore book it externally (Section 2.8). This convention is what makes the adoption channel *contractionary* under the price shock, since the cost lands in the periods household income is falling hardest (Section 4.1). The *ordering* of the policy results does not depend on it—it follows from the relative price alone—but the sign of the adoption channel's output contribution under a price shock does, and should be read as conditional on this booking.

3. **Zero pass-through to the green price** ($\rho_g = 0$) means adopters are completely insulated from the fossil shock: an upper bound on the adoption channel. A partial pass-through calibrated to observed electricity–gas co-movement would be more defensible.

4. **No modelled green technology.** The adopters' resource saving is maintained as an assumption (Section 2.8) rather than produced by a green sector with its own capacity, cost curve and investment. This is what makes the adoption benefit an upper bound and leaves the current-account treatment reduced-form.

5. **No within-country energy-share heterogeneity.** Bayer et al. (2026)'s central distributional mechanism is that poor households have higher energy shares; we have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.

6. **No investment margin.** Both Bayer et al. (2026) and Auclert et al. (2024) have richer supply sides. Its absence here likely makes the transfer look more expansionary, and the supply shock more inflationary, than it would be with capital to smooth the adjustment.

# A Summaries of the three reference papers

These summaries are in our own words, for internal reference. They are deliberately compact—roughly one page each—and emphasise the features that matter for our comparison in Section 5.

## A.1 Auclert et al. (2024), "Managing an Energy Shock"

Auclert et al. (2024) ask how an energy-importing economy should respond to an energy price shock, and whether household heterogeneity changes the answer. The model is a small open economy HANK built on the exchange-rate model of Auclert, Rognlie, Souchier and Straub (2021), extended with an energy good. There are three goods: a domestically produced home good, an imported foreign good, and imported energy. Households face idiosyncratic productivity risk, cannot insure it, and hold a single asset through a mutual fund; preferences are CRRA in a CES consumption basket that nests energy against a home/foreign bundle, with a low energy substitution elasticity ($\eta_E = 0.1$) and a total substitution index $\chi = 0.3$. Crucially, all households share the *same* basket and the *same* price index—the energy technology is fixed and homothetic. Wages are sticky with a real-wage-stabilisation motive; the monetary baseline is a rule that holds the real interest rate constant. Energy is supplied by price-taking foreign firms with an intertemporal extraction margin, so in the single-country equilibrium the world energy price is exogenous.

The analytical core is a decomposition of the output response into an expenditure-switching channel (positive) and a real-income channel (negative) scaled by the matrix of intertemporal MPCs. In the complete-markets representative-agent benchmark only expenditure switching survives, so the shock is expansionary; with realistic MPCs and low $\chi$ the real-income channel dominates and the shock is contractionary—their central result, and the one we inherit. A neutrality result holds at $\chi = 1$, where the two channels cancel. With real wage rigidity a wage–price spiral can appear, but the real wage still falls.

On policy, ARS study monetary tightening and three fiscal tools—energy subsidies, targeted transfers (indexed to counterfactual energy use), and untargeted transfers, all deficit-financed and repaid by a labour tax. For a country acting alone, subsidies are the most effective domestic stabiliser and tame inflation (they cap the price that drives the real-wage loss); transfers support demand but are more inflationary. The decisive twist is *cross-country*: monetary tightening has a positive externality (it lowers world energy demand and price), while subsidies have a large negative externality (coordinated subsidies make world energy demand nearly inelastic, spike the world price, and become self-defeating). Their policy recommendation is coordinated tightening plus transfers targeted to the poor, avoiding subsidies. They also show the impulse responses are not strongly nonlinear at empirically relevant shock sizes.

For our purposes the key features are: fixed homothetic energy technology, a single energy price and CPI, the constant-real-rate monetary rule, and the fact that the only "cost" of a subsidy in their world is a cross-country externality—there is no transition margin.

### A.2 Bayer et al. (2026), "Hicks in HANK" / "Redistribution Within and Across Borders"

Bayer et al. (2026) evaluate the fiscal response to the European energy crisis in a two-country HANK model of a monetary union, calibrated with Germany as "Home" (one-third of the union) and the rest of the area as "Foreign" (calibrated to Italy). The model is a medium-scale, two-asset HANK: households hold a liquid union-wide bond and illiquid capital, face idiosyncratic income risk, and have GHH preferences; there is an investment margin with capacity utilisation. Energy enters both production (a CES bundle with the capital–labour aggregate, elasticity 0.2) and household consumption (a CES bundle with the physical good, elasticity 0.1). A distinctive and, for them, central feature is within-country heterogeneity in the *energy share*: households are of high- or low-energy-intensity type, following a persistent Markov process correlated with income, so that poorer households spend a larger share on energy and suffer larger real-income losses. Monetary policy is a standard union-wide Taylor rule; each national fiscal authority taxes labour and profits and stabilises its debt.

The defining assumption is that the total union energy supply is *inelastic*: import capacity is fixed in the short run, so a fixed quantity is set exogenously and a common price clears the market. The crisis is a 20% cut in supply for six quarters, in response to which the energy price rises roughly fivefold, inflation rises about four points, and output falls around one percent, with consumption falling most for poor, high-energy households.

They compare two policies, each implemented in Home only so that spillovers can be measured: a subsidy that caps the retail energy price at its pre-crisis level, and a Hicks/Slutsky transfer indexed to pre-crisis energy consumption (to households and firms). The subsidy is the more effective domestic stabiliser but is fiscally expensive and *beggar-thy-neighbour*: by holding Home's energy demand up it pushes the common price higher, amplifying the crisis in Foreign. The transfer stabilises less at Home but generates essentially no spillover and is cheaper. On welfare (consumption-equivalent variation), the subsidy *reduces* Home welfare—subsidising energy under a supply contraction is inefficient because it suppresses both intratemporal and intertemporal substitution—while the transfer *raises* Home welfare, its cost being mostly the distortion of the taxes that finance it. Their headline contrast with ARS (subsidy more stabilising than transfer for output) they attribute to the Taylor rule and the two-country structure, versus ARS's constant-real-rate rule.

For our purposes the key features are: fixed union-wide quantity (our inelastic closure), the two-country spillover mechanism that we cannot represent, the within-country energy-share heterogeneity that we do not have, and the substitution-preserving property of the transfer, which we share.

### A.3 Langot et al. (2026), "The Macroeconomic and Redistributive Effects of the French Tariff Shield"

Langot et al. (2026) evaluate the French *bouclier tarifaire*—the 2022–23 cap on gas and electricity prices—in a HANK model of France as a single economy within the euro area. The model extends Auclert, Rognlie, Souchier and Straub with energy in both household consumption and production. Households face idiosyncratic productivity risk and hold a single asset; preferences are CRRA with a CES energy nest, and the energy substitution elasticity is $\eta_E = 0.5$, markedly higher than in ARS or Bayer. A distinctive feature is an *incompressible* (subsistence) level of energy consumption: only energy above the subsistence level yields utility. This generates non-homotheticity without non-homothetic preferences—poorer households devote a larger share to incompressible energy and have a lower price elasticity of energy demand, which is the source of the distributional results. The ECB Taylor rule is down-weighted to reflect France's small weight in euro-area inflation. In the policy simulations the energy price is taken as exogenous (from the government's forecasts), so France is effectively a price taker facing perfectly elastic supply.

The methodology is distinctive and worth flagging: rather than clean MIT-shock impulse responses, the paper performs an *ex-ante, conditional-forecast* evaluation. Using the sequence-space Jacobian, the authors back out the sequence of structural shocks (energy price, government spending, transfers, plus latent demand, markup and fiscal shocks) that makes the model reproduce the French government's official published forecasts for output, inflation and the debt ratio through 2027. Policies are then evaluated as counterfactuals that hold this shock sequence fixed and change one instrument.

The main finding is that the tariff shield supports growth (about 1.9% average over 2022–23 against 1.0% without it) and curbs inflation (5.6% against 7.2%), at a fiscal cost near 2% of GDP per year and a modest rise in the debt ratio, while containing—though not eliminating—the rise in consumption inequality even though it is untargeted. The mechanism behind its effectiveness is a *labour-cost* channel: by holding down the retail energy price the shield prevents the price–wage spiral, contains the cost of labour, and thereby supports employment and output. On this macro criterion the shield beats both a faster wage indexation (more inflationary, worse for output) and a targeted transfer to finance incompressible energy consumption; the targeted transfer, however, reduces inequality by more.

For our purposes the key features are: exogenous/elastic energy price (our price-taking closure), the labour-cost channel behind the pro-shield result (which our wage block only partly captures), the higher intensive substitution elasticity, and again a fixed energy technology with no transition margin.

## References

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